

Analog-to-Digital & Digital-to-Analog Conversion

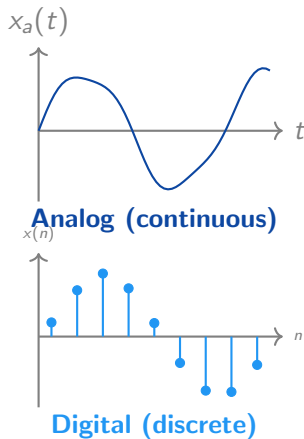
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Why Do We Need A/D Conversion?

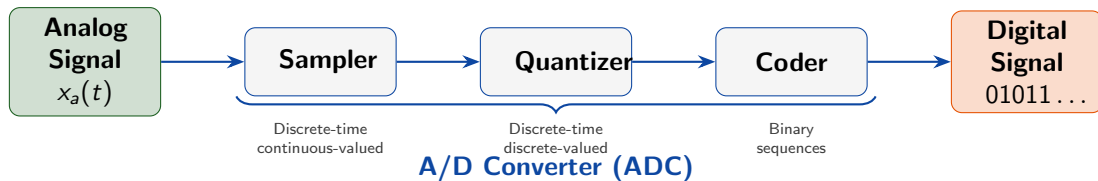
Real-world signals are analog

- Speech, music, ECG, seismic waves
- Radar & sonar signals
- Video & images
- All vary *continuously* in time and amplitude

Problem: Computers only work with **discrete numbers of finite precision.**



The A/D Conversion Pipeline



Key Insight

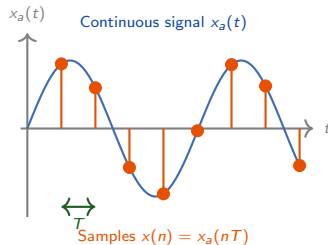
Although we model ADC as three separate steps, in practice a **single device** performs all three operations. Sampling *always* precedes quantization.

Definition

Convert a **continuous-time** signal $x_a(t)$ into a **discrete-time** signal $x(n)$ by measuring its value at equally spaced instants:

$$x_a(nT) = x(n), \quad n \in \mathbb{Z}$$

where T is the **sampling interval** (seconds).



- $f_s = \frac{1}{T}$ is the **sampling frequency** (Hz)
- Sampling is **reversible** — no information is lost if f_s is high enough
- Key condition: **Nyquist–Shannon Sampling Theorem**

$$f_s \geq 2 f_{\max}$$

Sampling — The Aliasing Problem

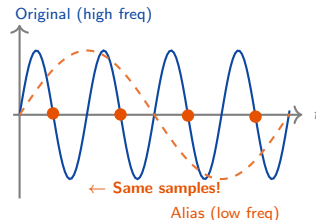
What is Aliasing?

If $f_s < 2f_{\max}$, high-frequency components **masquerade** as lower frequencies in the sampled signal — *this is aliasing*.

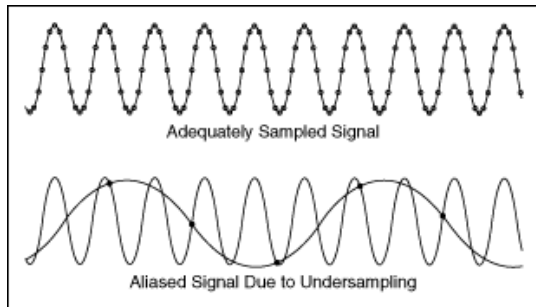
How to Avoid It

- Ensure $f_s \geq 2f_{\max}$ (Nyquist rate)
- Use an **anti-aliasing filter** before sampling to band-limit the signal

Example: Audio CD uses $f_s = 44.1$ kHz because human hearing reaches ≈ 20 kHz, so $2 \times 20 = 40$ kHz is the minimum required rate.



The Aliasing Problem



What is Aliasing?

Definition

When $f_s < 2f_{\max}$, high-frequency components **masquerade as lower frequencies** in the sampled signal. This is **aliasing**.

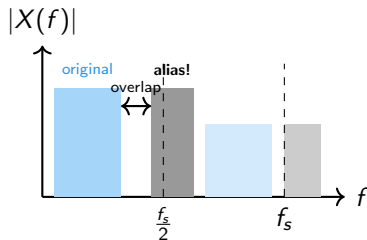
A signal at frequency f sampled at f_s *appears* to be at alias frequency:

$$f_{\text{alias}} = |f - k f_s|, \quad k \in \mathbb{Z}$$

Choose k such that $f_{\text{alias}} \in [0, \frac{f_s}{2}]$.

Example: $f = 900 \text{ Hz}$, $f_s = 1000 \text{ Hz}$

$$f_{\text{alias}} = |900 - 1 \times 1000| = 100 \text{ Hz}$$



The Nyquist–Shannon Sampling Theorem

Theorem (Nyquist–Shannon, 1949)

A band-limited signal with highest frequency $f_{\max}$ can be *perfectly* reconstructed from its samples **if and only if**:

$$f_s \geq 2 f_{\max}$$

The minimum rate $f_N = 2 f_{\max}$ is called the **Nyquist rate**.

In angular frequency:

$$\Omega_s \geq 2 \Omega_{\max}$$

where $\Omega_s = 2\pi f_s$ and $\Omega_{\max} = 2\pi f_{\max}$.

Example

Human hearing: $f_{\max} \approx 20$ kHz

CD audio: $f_s = 44.1$ kHz

$44.1 > 2 \times 20 = 40$ ✓

Example

Consider the analog signal

$$x_a(t) = 3 \cos 100\pi t$$

- (a) Determine the minimum sampling rate required to avoid aliasing.
- (b) Suppose that the signal is sampled at the rate $F_s = 200$ Hz. What is the discrete-time signal obtained after sampling?
- (c) Suppose that the signal is sampled at the rate $F_s = 75$ Hz. What is the discrete-time signal obtained after sampling?
- (d) What is the frequency $0 < F < \frac{F_s}{2}$ of a sinusoid that yields samples identical to those obtained in part (c)?

(a) Minimum Sampling Rate

Given the analog signal:

$$x_a(t) = 3 \cos(100\pi t)$$

Compare with:

$$x_a(t) = 3 \cos(2\pi Ft)$$

$$2\pi F = 100\pi \Rightarrow F = 50 \text{ Hz}$$

Nyquist criterion:

$$F_s \geq 2F = 100 \text{ Hz}$$

$$F_s^{\min} = 100 \text{ Hz}$$

(b) Sampling at $F_s = 200$ Hz

Discrete-time signal:

$$x[n] = x_a(nT_s) = 3 \cos \left(100\pi \cdot \frac{n}{200} \right)$$

$$x[n] = 3 \cos \left(\frac{\pi}{2} n \right)$$

$$x[n] = 3 \cos \left(\frac{\pi}{2} n \right)$$

Observation: No aliasing since $F_s > 2F$.

(c) Sampling at $F_s = 75$ Hz

$$x[n] = 3 \cos \left(100\pi \cdot \frac{n}{75} \right) = 3 \cos \left(\frac{4\pi}{3} n \right)$$

Reduce frequency to principal range:

$$\frac{4\pi}{3} - 2\pi = -\frac{2\pi}{3}$$

$$x[n] = 3 \cos \left(-\frac{2\pi}{3} n \right) = 3 \cos \left(\frac{2\pi}{3} n \right)$$

$$x[n] = 3 \cos \left(\frac{2\pi}{3} n \right)$$

Observation: Aliasing occurs since $F_s < 2F$.

(d) Aliased Frequency)

From part (c), the discrete-time frequency is:

$$\omega = \frac{2\pi}{3}$$

Relation between analog and discrete frequency:

$$\omega = 2\pi \frac{F}{F_s}$$

Solve for F :

$$F = \frac{\omega}{2\pi} F_s = \frac{2\pi/3}{2\pi} \times 75 = 25 \text{ Hz}$$

Interpretation:

- Original signal frequency = 50 Hz
- Sampling rate = 75 Hz (below Nyquist)
- The sampled signal appears as a 25 Hz signal

(d) Aliased Frequency

Aliasing concept:

$$F_{\text{alias}} = |F - kF_s|$$

$$|50 - 75| = 25 \text{ Hz}$$

Conclusion:

- 50 Hz and 25 Hz produce identical samples at $F_s = 75 \text{ Hz}$
- Hence, 50 Hz is an alias of 25 Hz

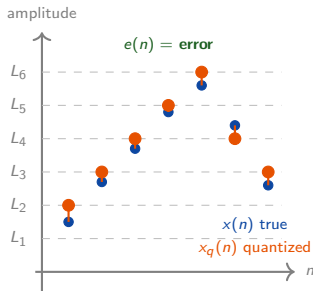
$$F = 25 \text{ Hz}$$

Definition

Map each sample $x(n)$ (which can be *any* real value) to the **nearest** value from a *finite* set of levels.

$$x_q(n) = Q[x(n)]$$

- With b bits $\Rightarrow 2^b$ quantization levels
- **Quantization error:** $e(n) = x(n) - x_q(n)$
- Quantization is **irreversible** (non-invertible)
- More bits $\Rightarrow$ smaller error $\Rightarrow$ higher cost



Common bit depths

8-bit	256 levels (telephone)
16-bit	65536 levels (audio CD)
24-bit	16.7M levels (studio)

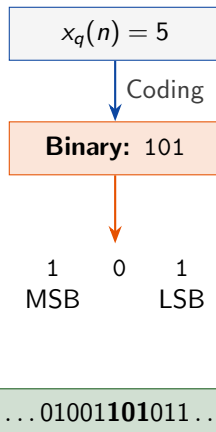
Definition

Represent each quantized value $x_q(n)$ as a **b -bit binary sequence**.

- Assigns a unique binary code to each quantization level
- With b bits: codes range from $\underbrace{00 \dots 0}_b$ to

$\underbrace{11 \dots 1}_b$

- Output is the stream $01011\dots$ fed to a digital processor



Example

Example: 3-bit coding (8 levels)

Level	Binary code
0	000
1	001
2	010
3	011
4	100
5	101
6	110
7	111

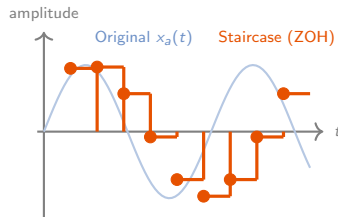
Digital-to-Analog (D/A) Conversion

Why D/A?

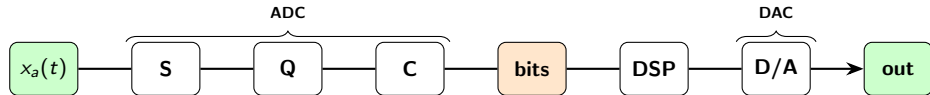
We cannot listen to numbers! After digital processing, signals must be converted *back* to analog form.

D/A converters “connect the dots” using interpolation:

- 1 **Zero-order hold** — staircase; hold each sample constant until the next
- 2 **Linear interpolation** — straight lines between successive samples
- 3 **Quadratic interpolation** — fit a parabola through 3 samples



Summary — Full Conversion Chain



Key Properties

- Sampling: **reversible** (if $f_s \geq 2f_{\max}$)
- Quantization: **irreversible** — always introduces distortion
- More bits $\Rightarrow$ less distortion, higher cost

Important Formulae

$$f_s \geq 2f_{\max} \quad (\text{Nyquist})$$

$$e(n) = x(n) - x_q(n) \quad (\text{Q. error})$$

$$\text{Levels} = 2^b \quad (b = \# \text{ bits})$$

Thank You!

Questions?

Remember: Sampling loses no info if $f_s \geq 2f_{\max}$
Quantization *a/ways* distorts — use more bits to reduce it.